Manuscript outline

Cell enlargement, polarity competition and migration persistence

Replacement resistance, conduction-limited response and stochastic polarity persistence

26 October 2026 · Round 7: cytosolic conduction and heading relocations

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes established polarity with local kinetics held fixed. Length represents enlargement. Published migration motivates this hypothesis; model heading and migration persistence remain distinct readouts.

Figures 1–2 retain the preceding evidence unchanged. Larger cells require longer fixed-width and proportional-width cues for replacement. Initiation, domain expansion and pulse-train retention distinguish the response mechanisms; censored results and nonmonotone deBelly responses remain qualified.

Figure 3 separates strong-cue turning from weak boundary response. The conduction test compares exact ring gains with an isolated-front prediction for Wave-pinning, inactive-diffusivity sweeps in Holmes and Spring, and the exceptional steep gain decline in Goryachev. Strong-cue turning times are not inferred from weak-input gain.

Figure 4 opens with drift and jump baselines, then shows heading correlation and the local size exponent across input strengths. Greater persistence of larger rings is tested against geometry rather than identified with cell movement.

Figure 5 compares measured drift with deterministic response predictions, tests drift-plus-takeover closure, and displays unfiltered heading-change distributions. Strong-input patch fields and native/fast-cytosol mode rates test whether long-range transfer contributes to persistence; seed uncertainty and discrepancies with the pilots remain explicit.

Figure 6 is the preceding migration Figure 5, renumbered with its scientific content unchanged. Published migration summaries support greater persistence in enlarged groups, with conditional OU estimates and visible model inadequacy.

Figure 7 reserves the SUM-159 paired WGD motility assays. No new experimental observations are available. Boundary photoactivation, full-perimeter activity and cytosolic mobility are prospective discriminating measurements, not completed assays.

Manuscript outline · continued

Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures
- Figure 3. Enlarged rings turn more slowly and respond less to weak boundary input
- Figure 4. Larger cells maintain heading longer under fluctuating input
- Figure 5. Physical drift, raw heading changes and cytosolic transport constrain the persistence mechanism
- Figure 6. Published migration summaries support a persistence hypothesis with conditional OU interpretation
- Figure 7. RESERVED: new WGD motility-assay results

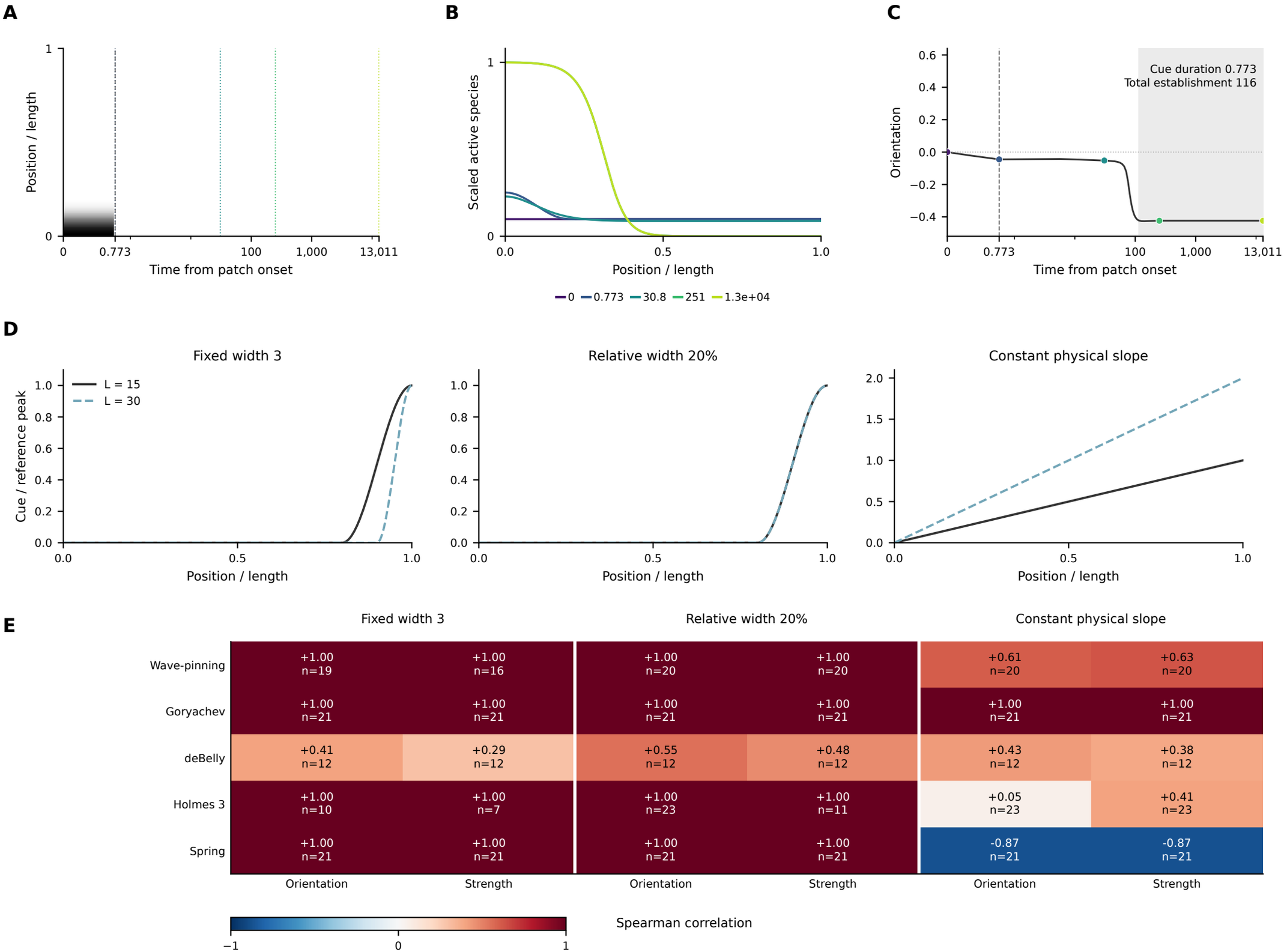

A
Position / length
Time from patch onset
0
0.773
100
1,000
13,011
B
Scaled active species
Position / length
0
0.773
30.8
251
1.3e+04
C
Orientation
Cue duration 0.773
Total establishment 116
Time from patch onset
D
Fixed width 3
Relative width 20%
Constant physical slope
Cue / reference peak
L = 15
L = 30
Position / length
E
Fixed width 3
Relative width 20%
Constant physical slope
Wave-pinning
Goryachev
deBelly
Holmes 3
Spring
+1.00 n=19
+1.00 n=16
+1.00 n=20
+1.00 n=20
+0.61 n=20
+0.63 n=20
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
+0.41 n=12
+0.29 n=12
+0.55 n=12
+0.48 n=12
+0.43 n=12
+0.38 n=12
+1.00 n=10
+1.00 n=7
+1.00 n=23
+1.00 n=11
+0.05 n=23
+0.41 n=23
+1.00 n=21
+1.00 n=21
+1.00 n=21
+1.00 n=21
-0.87 n=21
-0.87 n=21
Orientation
Strength
Spearman correlation

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Recorded Wave-pinning preparation cue at length 15: a finite left patch, withdrawn at 0.773 native time units. White to black is zero to peak input; coloured vertical lines mark the profile samples in B.

B, Native active profiles from the same preparation, scaled using the common minimum and span of the displayed snapshots. Colours identify recorded times.

C, Orientation from the same preparation. Grey marks basal verification after establishment arrival at about 116; the cue has already ended. The matched time scale in A and C separates input duration from autonomous development.

D, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

E, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

deBelly uses corrected endpoint commands. Spring retains native length and time units. Cross-model amplitudes and clocks are not calibrated biochemical doses or cellular times.

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

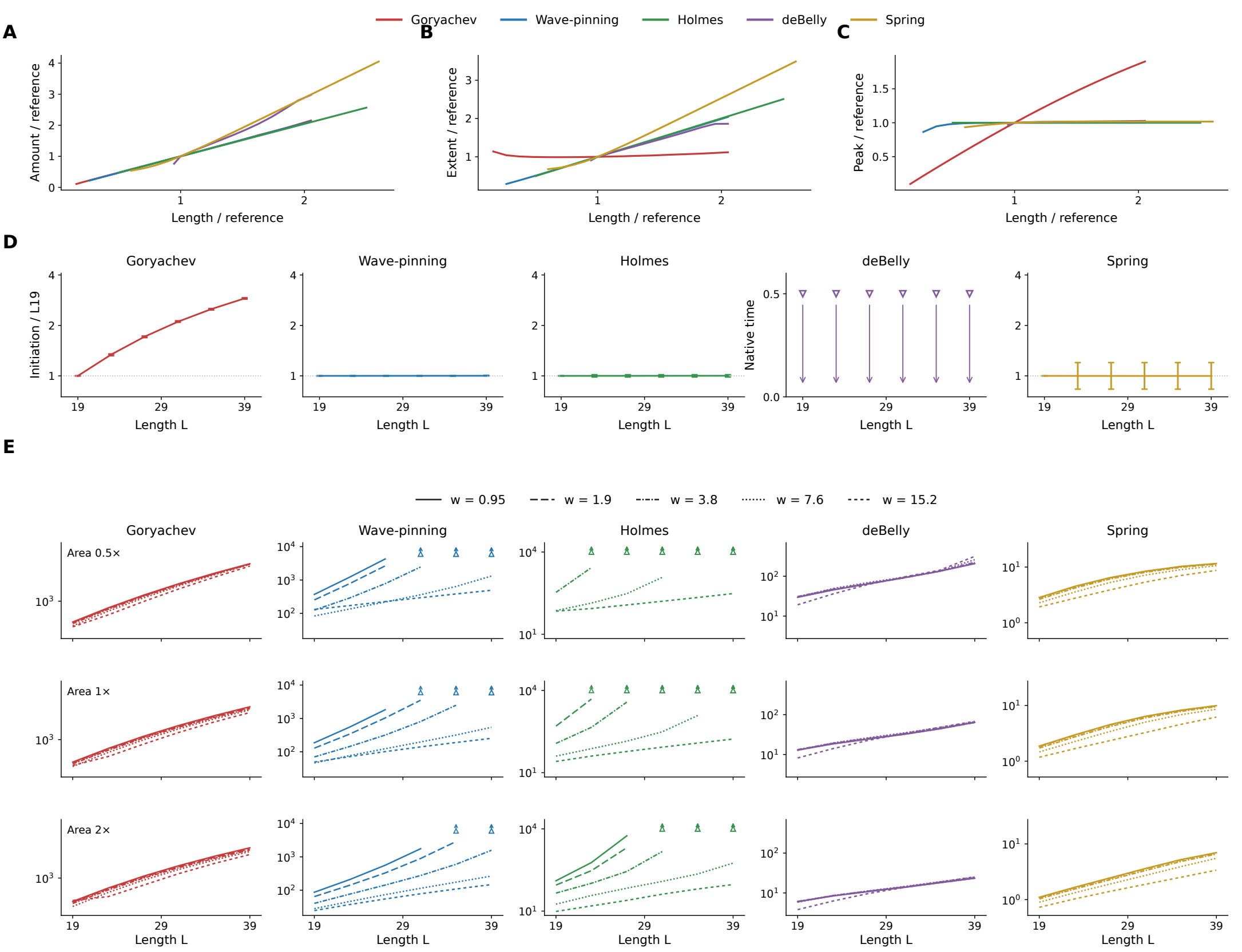
Sources and reproducibility

Three-test protocol and Spearman rule

Correlation table

Corrected endpoint method

Current duration curves


Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
B
Extent / reference
Length / reference
C
Peak / reference
Length / reference
D
Goryachev
Wave-pinning
Holmes
deBelly
Spring
Initiation / L19
Native time
Length L
E
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
Length L
F
Pulse-2 gain / headroom
Gap-end / pulse 1
Open circles: below activity gate at x = 0. Size increases from light (L19) to dark (L39).

G
Goryachev
lost
Wave-pinning
kept
Holmes
kept
deBelly
lost then commits
Spring
lost then commits
Challenger share
Time / pulse
L39
H
WGD−
WGD+
Initiation-limited
Goryachev
Saturating activation
Wave-pinning / Holmes
Slow non-activity state
deBelly / Spring
Biosensor
(schematic)
Time (schematic units)
I
Replacement time
(schematic)
narrow
broad
Width at fixed dose

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures

Separate size-dependent initiation and cue-width responses from observable pulse-memory classes.

Detailed legend

Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.

A, Incumbent excess amount normalized to the model reference, across the same 97 retained states.

B, Incumbent extent normalized to the model reference, for those same states.

C, Incumbent peak normalized to the model reference, for those same states.

D, Initiation is the time from cue onset until maximum in-cue activity minus the profile minimum exceeds 10% of the model's L19 incumbent excess peak. Area 1, width 3.8; times are normalized to L19. All six deBelly upper bounds are 0.5 native units (downward arrows); first-update sampling leaves its size dependence unresolved.

E, Replacement time (first eligible challenger share ≥ 0.75) versus length, with a line per width and rows for areas 0.5, 1 and 2. Upward triangles/arrows are lower bounds: the cue never replaced within the horizon (6,000 Wave-pinning; 10,000 Goryachev/Holmes; 2,000 deBelly; 100 Spring, native units). Broad cues rescue Holmes across all lengths; narrow-cue censoring does not establish a growing penalty. Spring native widths equal the displayed widths divided by 38.

F, Gap-1 activity kept versus pulse-2 gain as a fraction of headroom (Eq. 1). Four equal pulses, total ON 1.5 times the model-length commitment calibration; gaps 0–16 pulse durations. Holmes L31–39 uses the L27 calibration. Other models are grey; lightness indicates L. Open circles at x = 0 code activity below the existing visibility gate. At the left wall, deBelly/Spring gain on pulse 2 despite no visible activity; Goryachev never does. The deBelly L19/gap16 first-gap endpoint was checked against native fields.

G, L39 gap4 challenger-share histories, labelled kept / lost / lost then commits; shading marks cue ON. deBelly includes full recovery; others show one post-train pulse duration. Continuous raw shares include below-gate values. deBelly's later timing and fields remain numerically sensitive (supplement).

H, Prospective biosensor histories for the same pulse train in WGD− and enlarged WGD+ cells. Initiation-limited resetting (Goryachev), saturating activation with gap retention (Wave-pinning/Holmes), and progress retained after activity loss (deBelly/Spring) give distinct dimensionless schematic histories.

I, Prospective equal-dose width responses for those classes: weak width rescue, strong width rescue, and weaker width rescue. Solid grey: WGD−; dashed black: WGD+. These

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

are qualitative assay predictions.

Narrative points and conclusions

- At reference cue area 1, replacement is approximately 5–12-fold slower from L19 to L39 in every model wherever both lengths resolve. Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.
- The full width series shows that Holmes can replace at every tested length when the cue is broad enough; narrow-cue censoring cannot establish a growing width-penalty ratio.
- Pulse-2 headroom gain separates low visible retention with residual progress from the initiation-limited reset response. Saturating activation retains visible activity through gaps.

Proposed measurements and experimental follow-up

Measure first in-cue biosensor rise, then vary width at fixed integrated dose.

Measure pulse-1, gap-1 and pulse-2 challenger shares under the same pulse train, followed by eventual commitment.

Scope, limitations and pending validation

Commanded input areas and native clocks are not calibrated across models.

deBelly initiation is unresolved within the first native update; later-gap timing and endpoint fields remain numerically sensitive.

F codes known below-gate activity at x=0; the exact raw ratio remains in the plotted-value CSV.

Holmes large-length trains have no target-length commitment calibration; visible retention differs from eventual replacement.

The three classes describe selected deterministic regimes; WGD biosensor predictions are prospective.

Model definitions and readouts

$$x = \frac{s_{\mathrm{gap}\,1}}{s_1}, \qquad y = \frac{s_2 - s_1}{1 - s_1} \tag{1}$$

s_1 and s_2 are challenger shares at pulse ends 1 and 2; s_gap1 is the first-gap-end share. An observed endpoint below the existing visibility gate is coded as zero visible share in F; its raw value remains in the CSV.

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

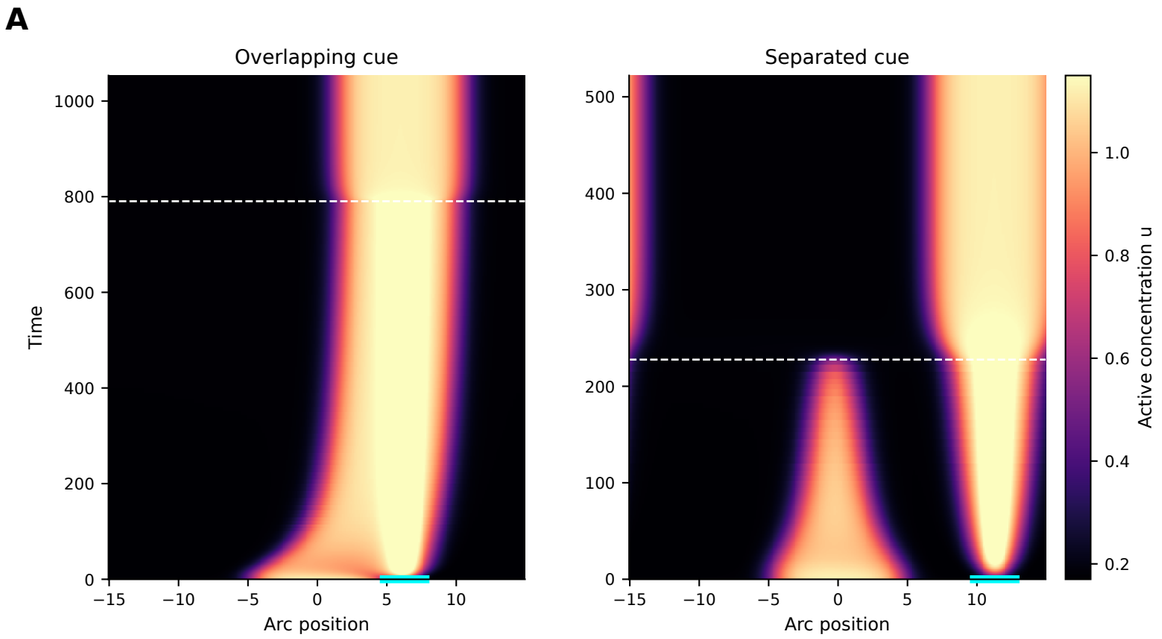
Sources and reproducibility

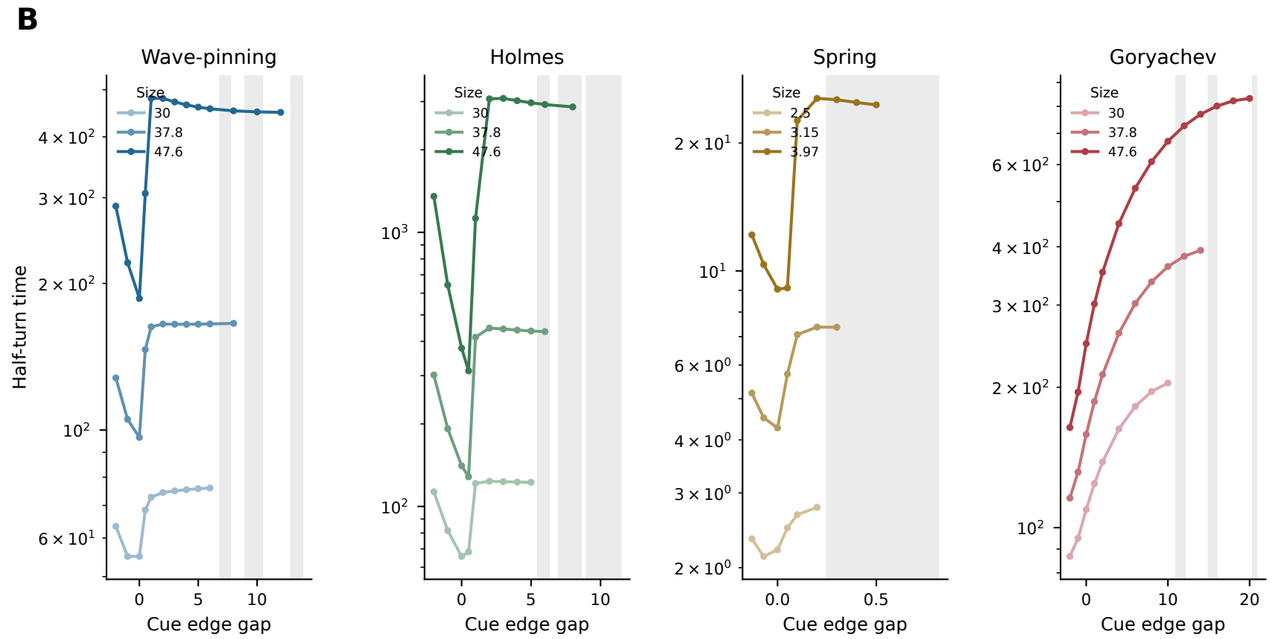
All 450 per-width replacement times

All 180 gap1/pulse2 coordinates and raw shares

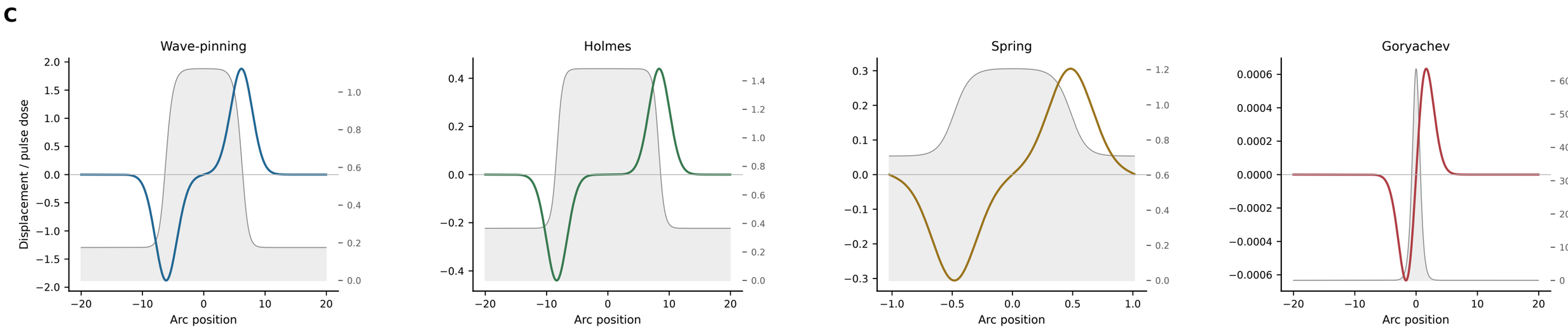
deBelly L19/gap16 native-phase audit

Preserved prior main panels and complete supplement

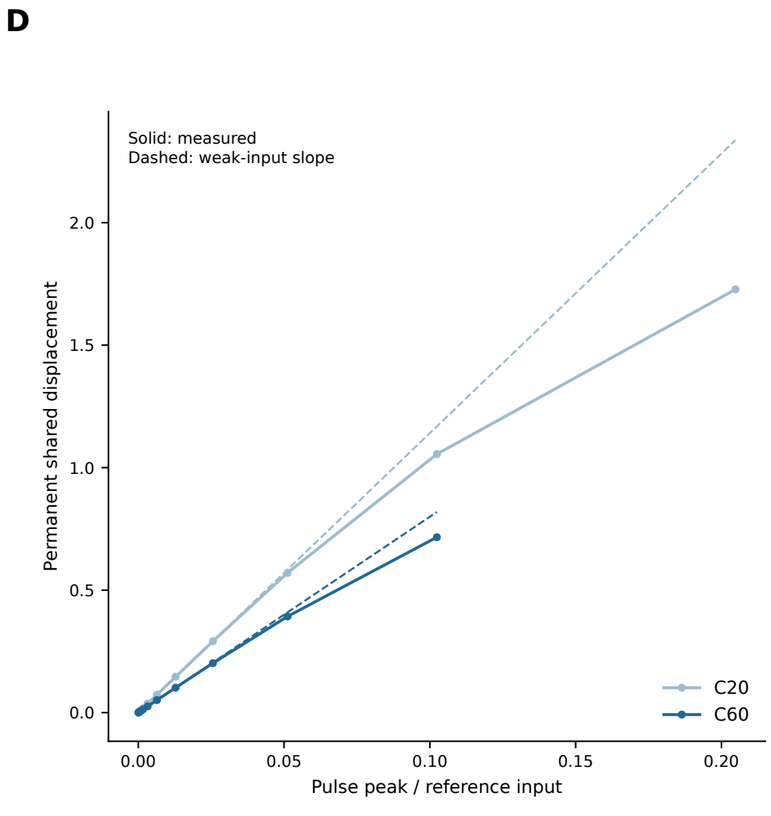




A: Wave-pinning C30, native continuous cue; dashed white: withdrawal. B: shades denote size; grey: heading undefined at ≥1 size.



C: f = 0.001, C40 (Spring 2). Colored line: permanent displacement per unit Gaussian-pulse dose; grey: mean profile.



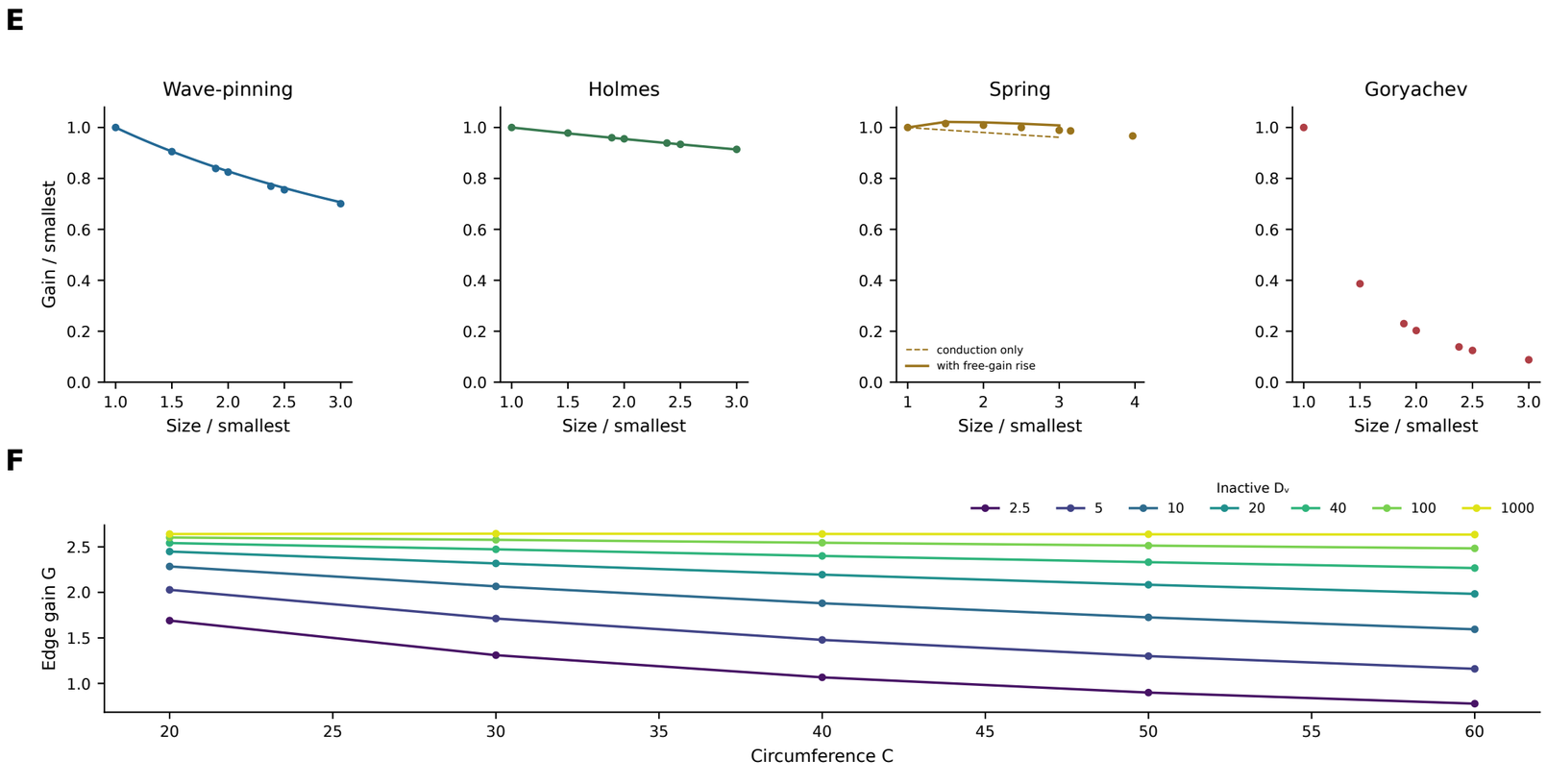


Figure 3. Enlarged rings turn more slowly and respond less to weak boundary input

Separates strong-cue turning from linear boundary susceptibility and tests the cytosolic conduction mechanism.

Detailed legend

Enlargement reduces boundary susceptibility, with a substantial cytosolic-conduction penalty in Wave-pinning and smaller native penalties in Holmes and Spring. Goryachev loses gain much more steeply than the conduction law predicts.

A, Native Wave-pinning C30 active-concentration kymographs for a cue overlapping the incumbent's 10%-contrast contour (edge gap −1) or separated from it (gap 4). Time starts at cue onset; cyan bars show spatial cue support. White dashed lines mark withdrawal after 95% of the target heading change; subsequent single-domain relaxation is included. Native cue peak 0.1 and support 3 are unchanged.

B, Time to turn halfway toward a continuous cue against its edge gap, at C30, 37.8 and 47.6 (Spring resting sizes 2.5, 3.15 and 3.97). Shades denote size within each model. Open triangles with upward arrows are right-censored lower bounds; grey spans mark gaps at which heading was undefined at one or more sizes. These strong-cue turning times are distinct from weak-pulse susceptibility.

C, Susceptibility to a weak Gaussian pulse, permanent displacement per peak×duration dose, against position at C40 (Spring resting size 2). Colored lines use the native translation-mode adjoint at f=0.001 (Spring mean input 0.004); grey curves and right axes show the same stationary mean active profile. The Gaussian SD is 1.5 (Spring 0.15). Signed gain is localized to the two lateral boundaries.

D, Wave-pinning shared permanent boundary displacement against edge-pulse peak divided by reference input 0.1, at C20 and C60; duration 50. Dashed lines use the weak-input gain. Ten-percent departures are bracketed between 5–20% of reference input in Wave-pinning, 2.6–5% in Holmes, 1.3–2.6% in Spring and 2.6–20% in Goryachev (neighboring tested amplitudes, pooled size ranges). Dose-matched duration sweeps agree within 0.2%.

E, Native edge gain normalized to each model's smallest size, against relative resting size. Lines show normalized $G=(\lambda/2)/(1+C/C^*)$. Wave-pinning $C^*=76.28$ comes from the independently solved isolated front, $\kappa=1.28149$, $\Delta u=0.96085$, and the stationary native domain fraction θ; no ring gain is fitted. Holmes $C^*=403.45$ comes from the native-D inverse-gain regression $1/G=a+bC$, $C^*=a/b$, corroborated by the diffusivity sweep. Spring transport $C^*=49.00$ comes from a through-origin free-conduction penalty regression $G(1000,C)/G(D,C)-1$ against C/D over $D\geq1$. Its dashed curve is the pure conduction law; its solid curve additionally includes the independently measured $G(1000,C)/G(1000,C_{min})$ rise. The apparent native inverse-gain crossover is 142.30, which reflects cancellation by this intrinsic rise and is not equated with the transport crossover. Holmes's alternative penalty fit gives 469.06. Goryachev has no conduction-law curve. Spring's actual circumference includes its small global extension.

Figure 3. Enlarged rings turn more slowly and respond less to weak boundary input · continued

The E native points retain round6 mean input 0.004; the Spring diffusivity sweep uses pilot mean 0.0004.

F, Wave-pinning adjoint edge gain against circumference at inactive diffusivities 2.5, 5, 10 (native), 20, 40, 100 and 1000. Every point uses the same weak mean input and unchanged reaction kinetics. All 35 Wave-pinning preparations retain a polarized branch, including D_v=2.5 at C60 where the pilot reported domain loss. Fast cytosol largely removes the native size dependence. All quantities use model-native units; none is a calibrated migration velocity or biochemical dose.

Narrative points and conclusions

- Larger rings take longer to turn toward a continuous cue across the tested positions. Overlap distinguishes incumbent sliding from nucleation and replacement by a separate domain; further separation has relatively little effect outside Goryachev.
- Weak-input translation susceptibility is localized to the two domain boundaries. Its size dependence is large in Wave-pinning and Goryachev and small in Holmes and Spring.
- An isolated-front conduction law accounts for Wave-pinning gain without fitting ring gains. Inactive-diffusivity sweeps extend the same physical test to Holmes and Spring; branch loss and deviations remain explicit.

Proposed measurements and experimental follow-up

Photoactivate at offsets spanning the incumbent boundary and measure half-turn time and sliding versus replacement.

Apply a weak boundary pulse and measure permanent common displacement of both lateral boundaries; distinguish boundary response from migration speed.

Scope, limitations and pending validation

Strong turning and weak response use different amplitudes and size grids; susceptibility does not explain the strong turning times.

Spring’s low-diffusivity branch changes lie outside the reported conduction law. Goryachev has no fitted conduction curve.

Model time, input and size have no cross-model biological calibration.

Model definitions and readouts

$$G = \frac{\lambda/2}{1 + C/C^*}, \qquad C^* = \frac{2D_v}{\kappa \Delta u \theta (1-\theta)} \tag{3.1}$$

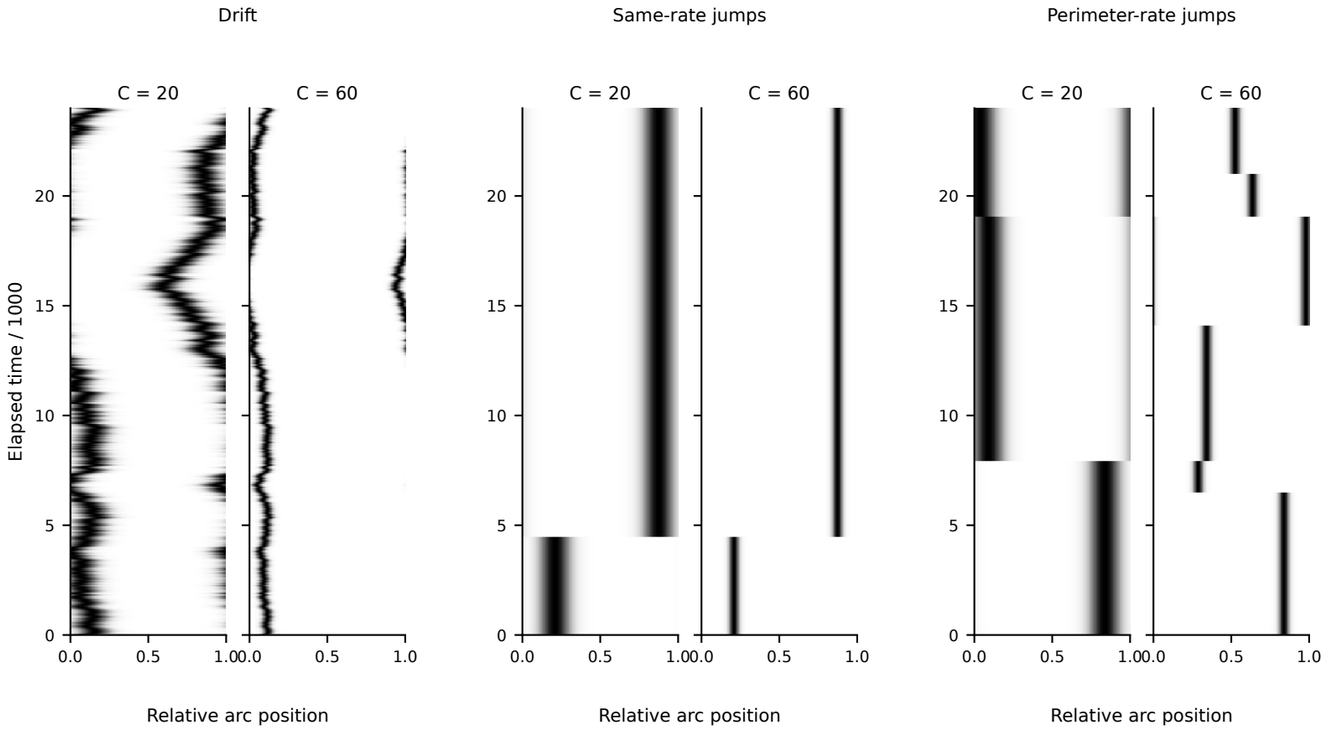
Figure 3. Enlarged rings turn more slowly and respond less to weak boundary input · continued

Wave-pinning crossover uses the isolated front and stationary domain fraction; Holmes uses the native inverse-gain regression and Spring uses the diffusivity-penalty regression, with intrinsic free-conduction rise shown explicitly.

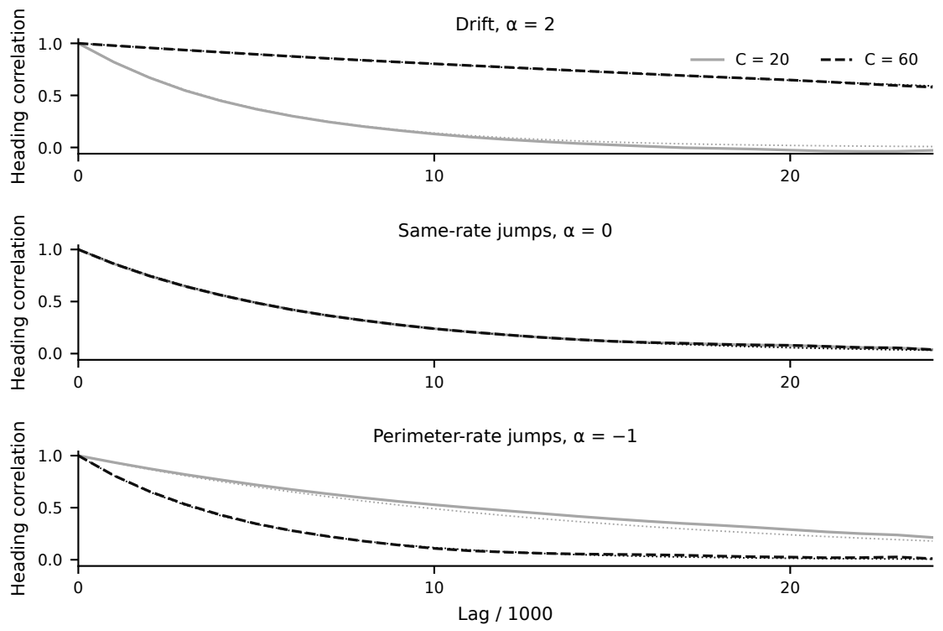
Sources and reproducibility

Scientific methods, departures from pilots and numerical scope
Diffusivity sweep and gains
Front quadrature and fitted crossover parameters

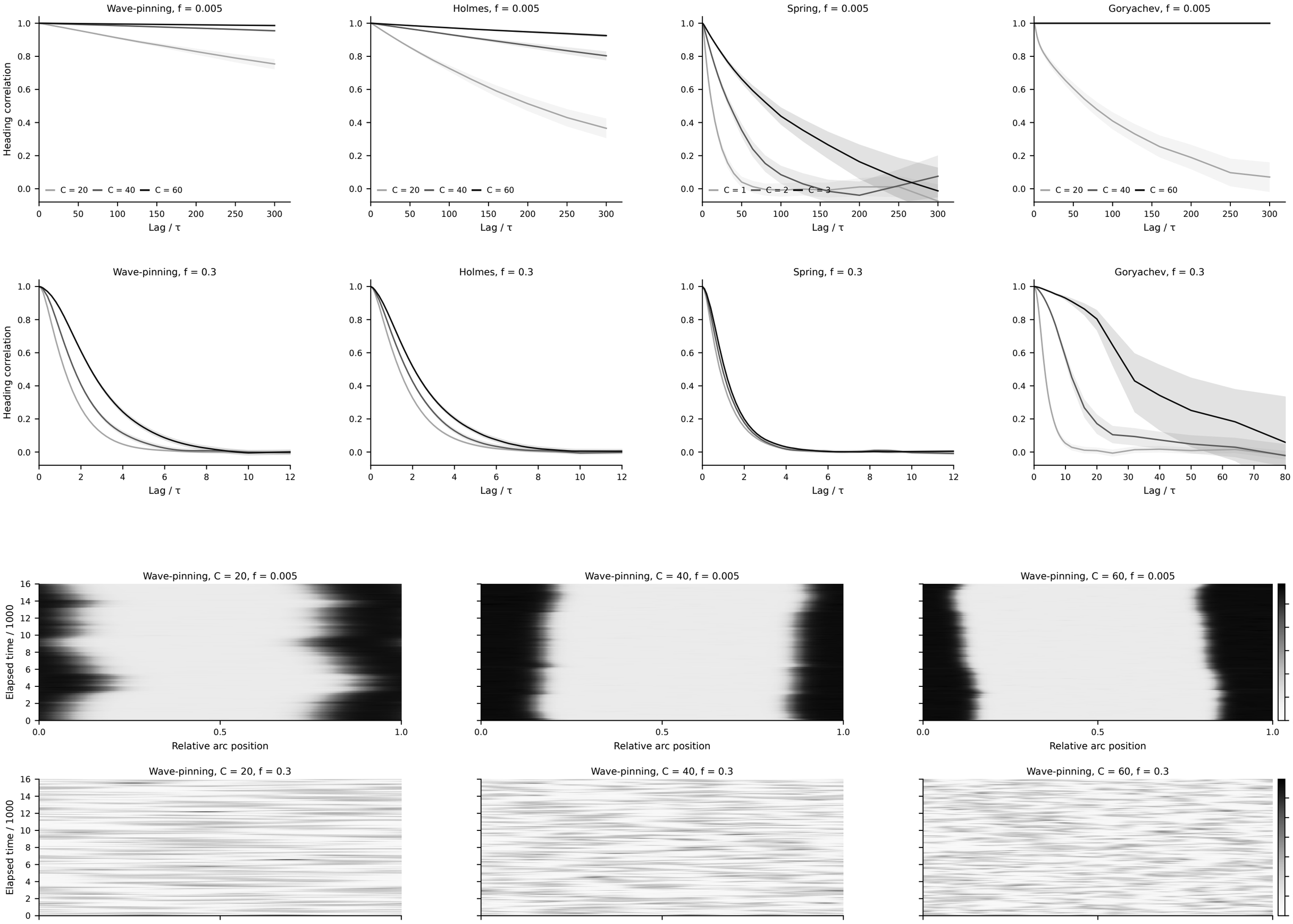
A


Drift
Same-rate jumps
Perimeter-rate jumps
C = 20
C = 60
Elapsed time / 1000
Relative arc position

B


Drift, α = 2
Same-rate jumps, α = 0
Perimeter-rate jumps, α = −1
C = 20
C = 60
Heading correlation
Lag / 1000

C


Wave-pinning, f = 0.005
Holmes, f = 0.005
Spring, f = 0.005
Goryachev, f = 0.005
Wave-pinning, f = 0.3
Holmes, f = 0.3
Spring, f = 0.3
Goryachev, f = 0.3
C = 20
C = 40
C = 60
C = 1
C = 2
C = 3
Heading correlation
Lag / τ

D

Wave-pinning, C = 20, f = 0.005
Wave-pinning, C = 40, f = 0.005
Wave-pinning, C = 60, f = 0.005
Wave-pinning, C = 20, f = 0.3
Wave-pinning, C = 40, f = 0.3
Wave-pinning, C = 60, f = 0.3
Elapsed time / 1000
Relative arc position
Activity u

E

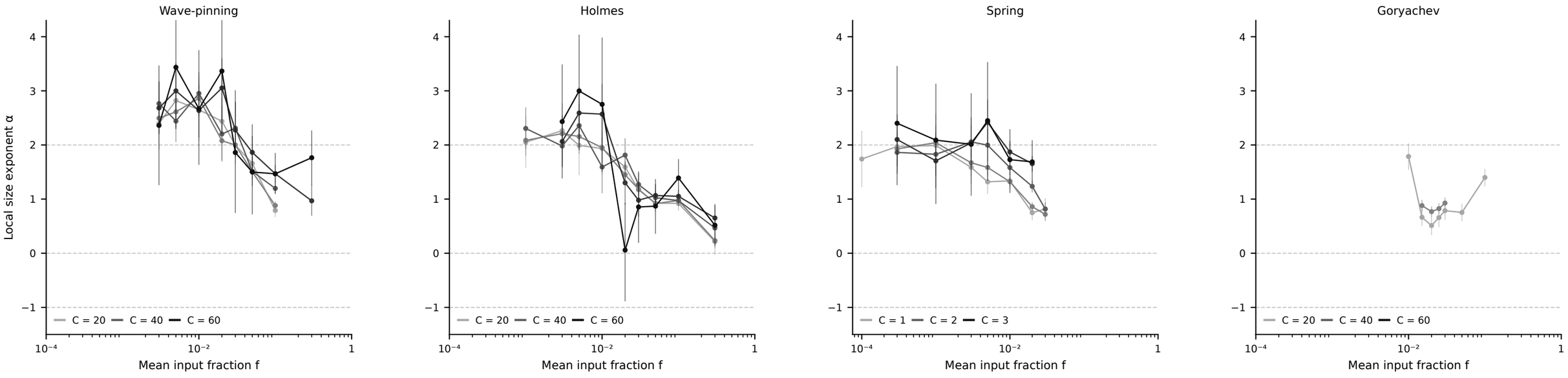

Wave-pinning
Holmes
Spring
Goryachev
C = 20
C = 40
C = 60
C = 1
C = 2
C = 3
Local size exponent α
Mean input fraction f

Figure 4. Larger cells maintain heading longer under fluctuating input

Native polarity-heading persistence and mechanistic comparisons

Detailed legend

Larger native rings retain heading longer under fluctuating input, with the largest size exponents in the weak-input regime. The baseline examples distinguish geometric persistence from a size-dependent change in physical dynamics.

A, Synthetic activity kymographs for a single drifting peak, same-rate random relocations and relocations whose rate scales with circumference, at C20 and C60. Position is relative arc length; grayscale denotes activity from 0 (white) to 1 (black). Time increases upward. These pedagogical baselines have no biochemical kinetics.

B, Heading correlation in the same three baselines, from 1000 seeds; solid and dashed lines indicate C20 and C60, dotted lines the analytic curves. A fixed physical drift diffusivity gives k proportional to C^{-2} ($\alpha = 2$); size-independent jumps give $\alpha = 0$, and jumps per unit perimeter give $\alpha = -1$. The same-rate curves coincide.

C, Native-model heading correlation at common weak f = 0.005 and strong f = 0.3, for C20, C40 and C60 (Spring 1, 2 and 3). Size is a light-to-dark grayscale ramp within each facet. Shading is the 95% confidence interval from 2000 seed-bootstrap resamples of the complete retained native histories, recomputed on a dense early-lag grid. Weak-input facets show 0–320τ; strong-input facets show 0–12τ, Goryachev 0–80τ, to resolve their different correlation scales. τ = 50 native time units, Spring 5. Heading correlation is the mean cosine of the change in arg(P), where P is the first Fourier activity moment divided by total activity. These retained round6 estimates require $|P| \geq 0.1$ at both lag endpoints; their coverage is in condition_lag_metrics.csv. They differ from the ungated distributions in Figure5D.

D, Native Wave-pinning examples at f = 0.005 and 0.3 and C20, C40 and C60, selected as replicate 0 without a seed search. Every panel shows the entire retained post-burn-in interval 16000–32000, on relative arc position. Activity scales are common across sizes at each input; the color bars state the separate weak- and strong-input ranges. Weak input mainly shifts a coherent domain; strong input produces a changing patch field. The kymographs are illustrative, not an event classification.

E, Local α, the negative fitted slope of log heading-decorrelation rate against log circumference in adjacent-size neighborhoods, against mean input f. Lines are size neighborhoods centered on C20, C30, C40, C50 and C60 (Spring 1, 1.5, 2, 2.5 and 3); light to dark follows size. Vertical bars are seed-bootstrap 95% CIs. Missing estimates interrupt curves. Horizontal guides are $\alpha = 2$, 0 and -1. Native f multiplies reference input 0.1, 1, 4 and 6.4 in Wave-pinning, Holmes, Spring and Goryachev respectively; doses and clocks are not calibrated across models.

All simulations reuse the complete, frozen round6 compact campaign (96–384 independent seeds per condition). Local kinetics, basal preparations and solver outputs

Figure 4. Larger cells maintain heading longer under fluctuating input · continued

are unchanged. α and activity-heading persistence are model observables, distinct from migration persistence or fitted OU velocity-memory parameters.

Narrative points and conclusions

- Geometry alone increases heading persistence when a domain has unchanged physical drift. Size-independent random jumps do not provide that increase.
- At common weak and strong inputs, the displayed native heading correlations decay more slowly for larger rings, conditional on the stated coherence eligibility.
- Weak-input size exponents are near the drift baseline in Holmes and Spring and exceed it in Wave-pinning where resolved. The advantage narrows as input strengthens; unavailable estimates and their reasons remain in the compact source tables.

Proposed measurements and experimental follow-up

Proposed cellular assays measure full-perimeter polarity activity, coherence, boundary displacement and cytosolic-pool mobility in matched WGD− and WGD+ cells. These assays are prospective; no experiment is inferred from the model trajectories.

Scope, limitations and pending validation

The common f values compare model regimes, not matched biochemical receptor stimulation; native clocks and Spring length units remain model specific.

Autocorrelation in C uses the original $|P| \geq 0.1$ endpoint eligibility mask and is conditional on retained coherence; Figure5D exposes raw headings without that mask.

Kymographs do not establish that every heading change is a discrete relocation. Strong-input patch fields can have small first-mode coherence.

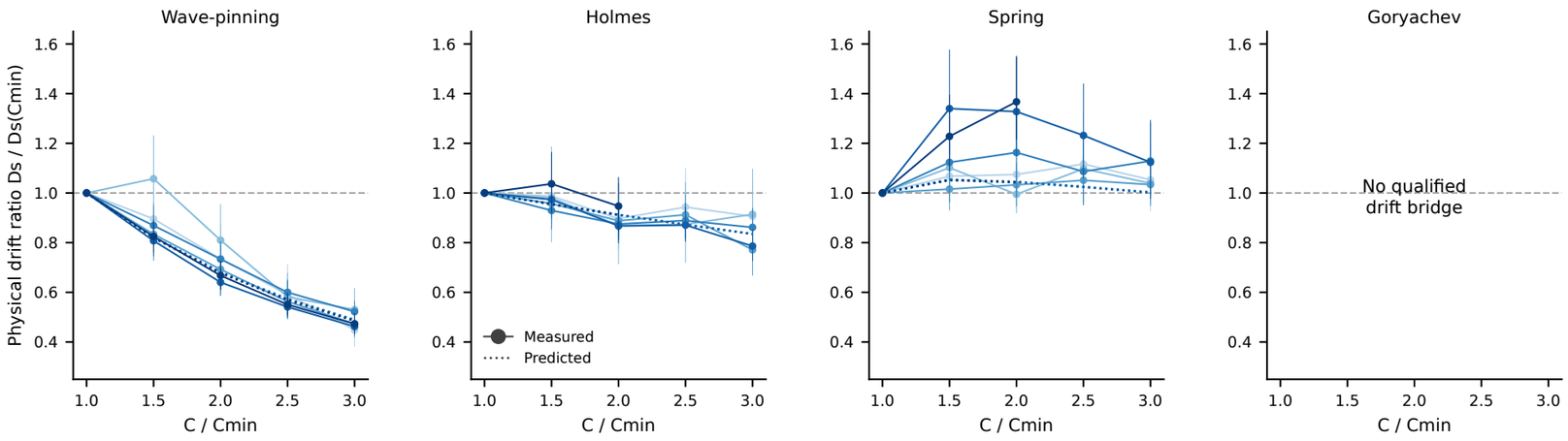
Seed-bootstrap intervals reflect the simulated seed ensemble and do not calibrate biological variation.

Sources and reproducibility

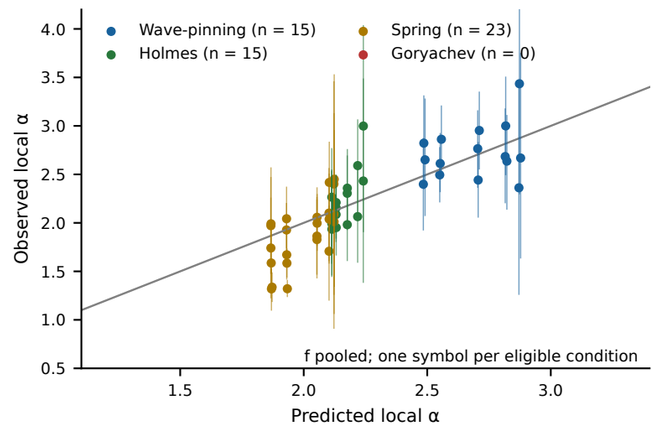
Main source data and definitions

Exact source hashes and generation time

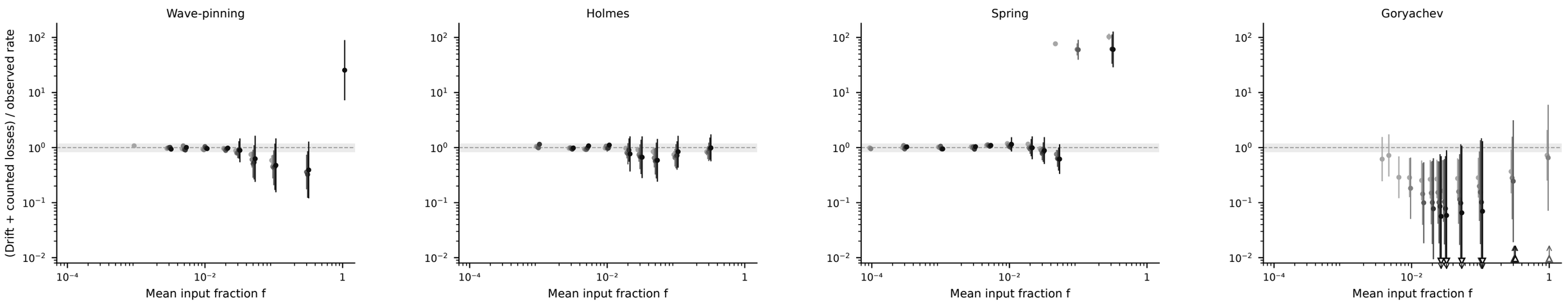
A


Wave-pinning
Holmes
Spring
Goryachev
Physical drift ratio Ds / Ds(Cmin)
C / Cmin
Measured
Predicted
No qualified
drift bridge

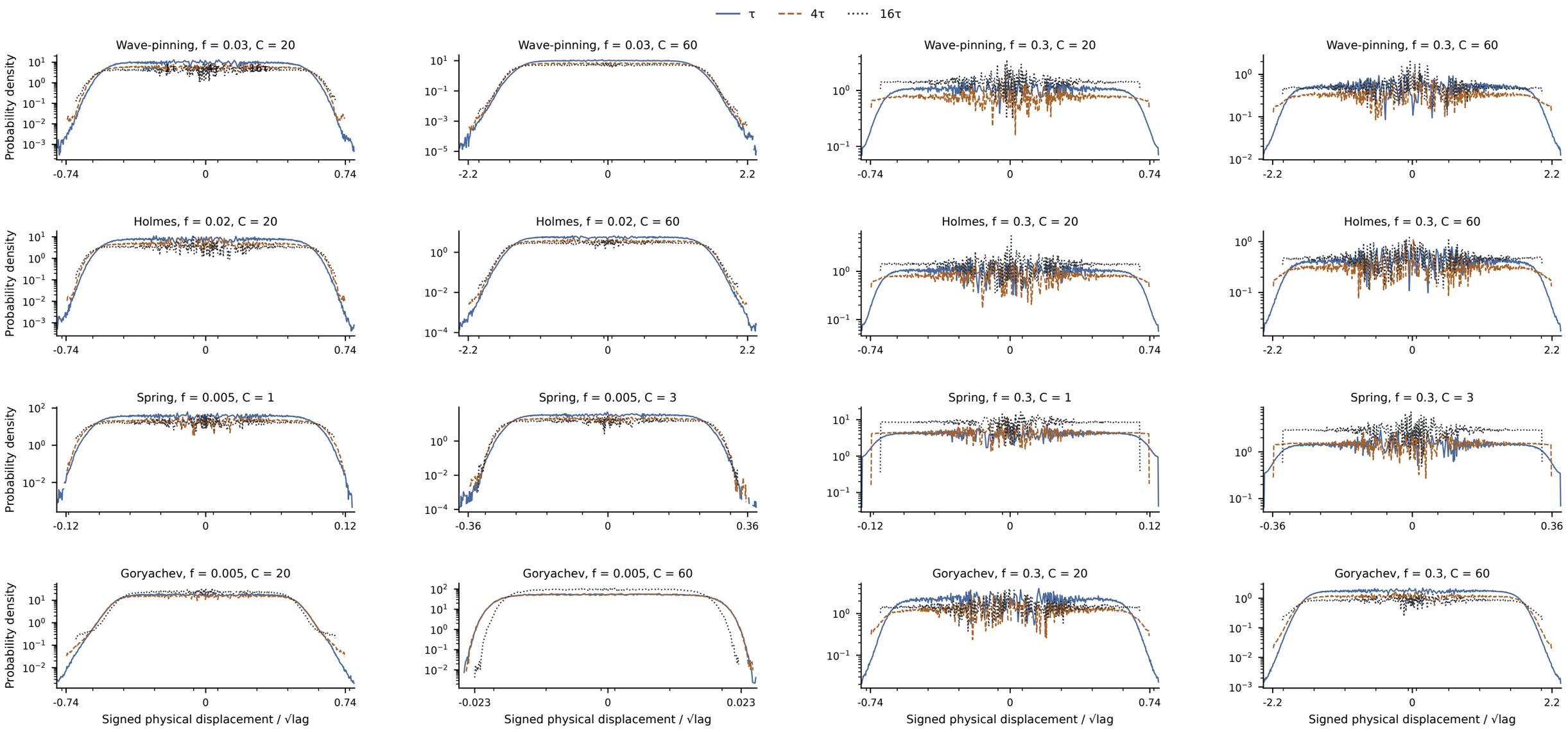
B


Wave-pinning (n = 15)
Holmes (n = 15)
Spring (n = 23)
Goryachev (n = 0)
Observed local α
Predicted local α
f pooled; one symbol per eligible condition

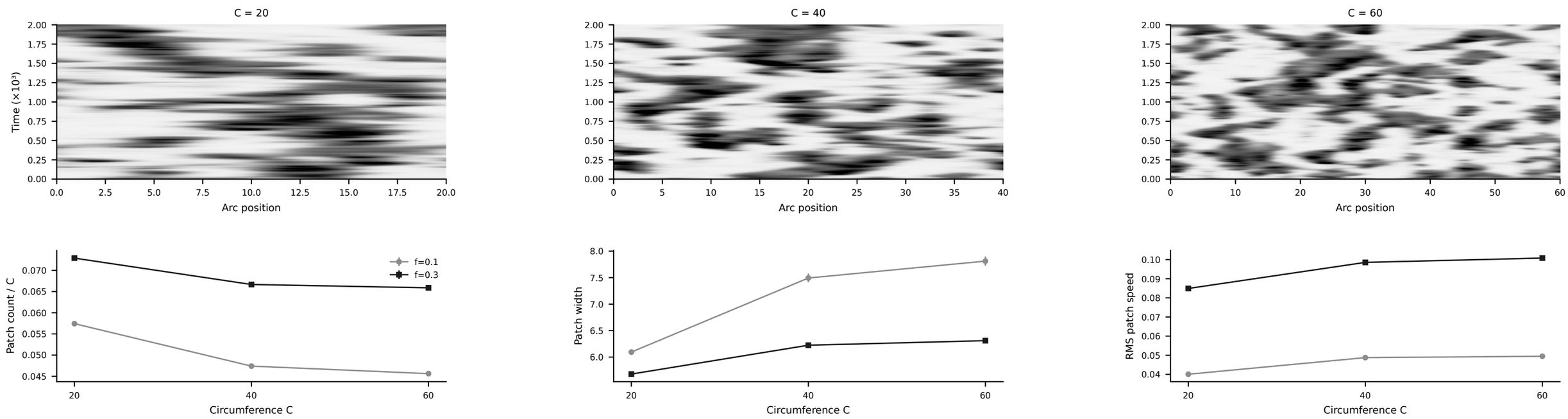
C


Wave-pinning
Holmes
Spring
Goryachev
(Drift + counted losses) / observed rate
Mean input fraction f

D


τ
4τ
16τ
Wave-pinning, f = 0.03, C = 20
Wave-pinning, f = 0.03, C = 60
Wave-pinning, f = 0.3, C = 20
Wave-pinning, f = 0.3, C = 60
Holmes, f = 0.02, C = 20
Holmes, f = 0.02, C = 60
Holmes, f = 0.3, C = 20
Holmes, f = 0.3, C = 60
Spring, f = 0.005, C = 1
Spring, f = 0.005, C = 3
Spring, f = 0.3, C = 1
Spring, f = 0.3, C = 3
Goryachev, f = 0.005, C = 20
Goryachev, f = 0.005, C = 60
Goryachev, f = 0.3, C = 20
Goryachev, f = 0.3, C = 60
Probability density
Signed physical displacement / √lag

E


C = 20
C = 40
C = 60
Time (×10³)
Arc position
Patch count / C
Patch width
RMS patch speed
f=0.1
f=0.3
Circumference C

F

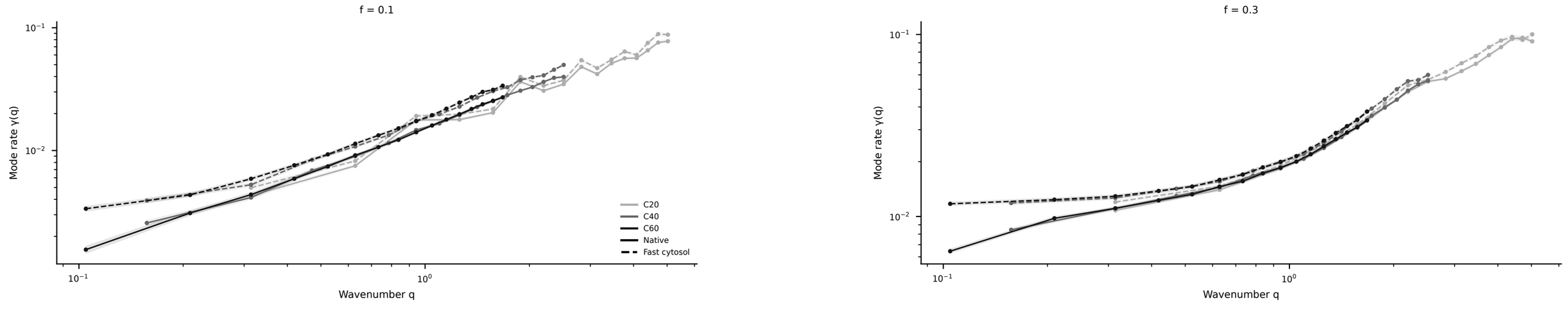

f = 0.1
f = 0.3
Mode rate γ(q)
Wavenumber q
C20
C40
C60
Native
Fast cytosol

Figure 5. Physical drift, raw heading changes and cytosolic transport constrain the persistence mechanism

Native polarity-heading persistence and mechanistic comparisons

Detailed legend

Weak-input physical drift is suppressed beyond the geometric baseline in Wave-pinning, while drift predictions fail quantitatively for Goryachev. Raw heading distributions and strong-input patch statistics expose the limits of treating every change as translation of one coherent domain.

A, Physical drift diffusivity Ds relative to the smallest ring at the same f, against relative circumference. Solid curves and circles are retained native trajectory estimates; dotted curves are the deterministic susceptibility prediction with shared translation and full input covariance. All designed f ≤ 0.01 with multiple-domain fraction ≤ 0.05 enter; color intensity indexes f (10^{-4}–10^{-2}). Vertical bars are retained seed-bootstrap 95% CIs. The geometric null is 1. Goryachev has no qualified drift bridge and is explicitly unavailable here, rather than treated as a drifting domain.

B, Predicted versus observed local α, pooled over eligible f ≤ 0.01, with a single circle symbol throughout and model color. Prediction comes from deterministic native susceptibility and the same adjacent-size neighborhoods as the observed α; no noisy-data fitting or pooling average is applied. Vertical bars are observed seed-bootstrap 95% CIs; the diagonal is equality. Eligibility requires a defined native mean state, linear-response validation, resolved observed α and neighborhood multiple-domain fraction ≤ 0.05. Goryachev contributes no eligible point, a quantitative failure retained in B_alpha_bridge.csv.

C, Drift plus counted takeover angular loss divided by observed heading decorrelation, at every designed input and size. For each counted takeover whose angle is unavailable, 1 − cos(Δθ) is bounded independently in [0,2]. Vertical segments span the resulting partial-identification interval, not a confidence interval; circles mark its geometric midpoint when positive finite endpoints exist. Gray band is 0.85–1.15. Open triangles with arrows show one-sided intervals or portions extending past the displayed range. Light to dark denotes C20–C60 (Spring 1–3). Missing observed rates cannot supply a ratio. If drift is missing its nonnegative lower bound can be used, but the upper limit is unbounded.

D, Raw heading-change probability density at lags τ (blue solid), 4τ (brown dashed) and 16τ (black dotted), with no coherence, contrast or track-identity cut. Weak input near the C20 transition is f = 0.03, 0.02, 0.005 and 0.005 in Wave-pinning, Holmes, Spring and Goryachev; strong input is f = 0.3 for all. Facets separate model, input and C20/C60 (Spring 1/3). Signed Δθ is the principal difference in arg(P), and physical displacement is mean endpoint circumference × Δθ/(2π). Binning is in physical displacement before x and probability density are transformed by 1/√lag and √lag respectively. The signed x axis is symmetric-logarithmic to reveal the narrow core and broad tails; y is logarithmic. τ = 50,

Figure 5. Physical drift, raw heading changes and cytosolic transport constrain the persistence mechanism · continued

Spring 5, in native time units. Zero-count bins are omitted without a pseudocount. All retained 96–384 independent seeds per condition enter. Collapse is descriptive, not a test passed by the data; broad tails can also reflect weak first-mode coherence or overlapping relocations.

E, Strong native input produces a Wave-pinning patch field with approximately extensive patch count and residual size dependence in local statistics. Top facets show the first 2000 recorded time units after burn-in at input fraction 0.3, C20/C40/C60, seed 0; grayscale is native active concentration from 0 to 1.5. Lower facets show mean patch density (count divided by circumference), mean physical patch width, and RMS physical patch speed at input fractions 0.1 and 0.3. Patches exceed active concentration 0.65 and span at least 10 native grid cells. Speed is centroid displacement divided by the 5-unit frame interval, between mutually best overlapping regions; unmatched and ambiguous comparisons are excluded. Points are means across 64 independent seeds with 95% seed-bootstrap intervals; widths and matched speeds first pool samples within each seed. Checks at thresholds 0.55 and 0.75 retain the observed directions of size dependence and are documented in the scientific readme. No birth or death statistic is inferred.

F, Faster cytosolic diffusion changes the low-wavenumber Wave-pinning mode rates. The plotted rate $\gamma(q)$ comes from each seed's normalized autocorrelation of the complex Fourier coefficient $a_n(t)$, defined below, at the sampled lag nearest correlation 0.5; $q = 2\pi n/C$. Modes 1–16 are shown when every seed resolves a rate beyond the 5-unit frame interval. Native inactive diffusivity 10 is solid; fast diffusivity 1000 is dashed; C20/40/60 follow a light-to-dark ramp. Each point is the geometric mean across 64 seeds; shading gives 95% percentile seed-bootstrap intervals. Open upward triangles with arrows mark first-lag resolution bounds. Fast-cytosol runs alter only inactive diffusivity and are mechanism tests. The native-minus-fast mode-1 size exponent is 0.544 [0.500, 0.592] at input fraction 0.1 and 0.438 [0.425, 0.452] at 0.3; brackets are 95% paired-seed bootstrap intervals across the three sizes. The Holmes pilot check gives smaller but resolved differences, 0.081 [0.046, 0.117] and 0.081 [0.076, 0.086], respectively.

Physical polarity drift, counted track switching, raw activity heading, complex Fourier-mode relaxation and migration velocity memory are distinct quantities. Fast inactive diffusion is a mechanistic perturbation; all native comparisons retain native local kinetics.

Narrative points and conclusions

- Wave-pinning drift falls below unchanged-physical-drift geometry and broadly follows deterministic susceptibility. Holmes and Spring sit closer to size-independent physical drift; Goryachev has no qualified drift bridge.

Figure 5. Physical drift, raw heading changes and cytosolic transport constrain the persistence mechanism · continued

- Raw heading distributions are presented at three lags without a coherence cut. Their interpretation is constrained by broad low-coherence phase changes and by lags that overlap counted relocations.
- Strong-input patch-field and mode-rate experiments test inactive-pool transport without changing the native main-comparison kinetics.

Proposed measurements and experimental follow-up

Proposed cellular assays measure full-perimeter polarity activity, coherence, boundary displacement and cytosolic-pool mobility in matched WGD− and WGD+ cells. These assays are prospective; no experiment is inferred from the model trajectories.

Scope, limitations and pending validation

Collapsing lag-rescaled distributions is a descriptive drift signature; a noncollapsing tail is not uniquely a track relocation, especially when |P| is small.

The largest lag can be comparable with or exceed the typical counted-switch interval. Zero detected switches do not prove a long biological waiting time.

C contains detector-conditioned counted events and partial-identification intervals; it is not an exhaustive causal decomposition or a confidence bound.

Native model reference doses, length units and time units are not experimentally calibrated across models.

Model definitions and readouts

$$a_n(t) = \frac{1}{C}\int_0^C u(s,t)e^{-iq_n s}\,\mathrm{d}s, \qquad q_n = \frac{2\pi n}{C} \tag{5.1}$$

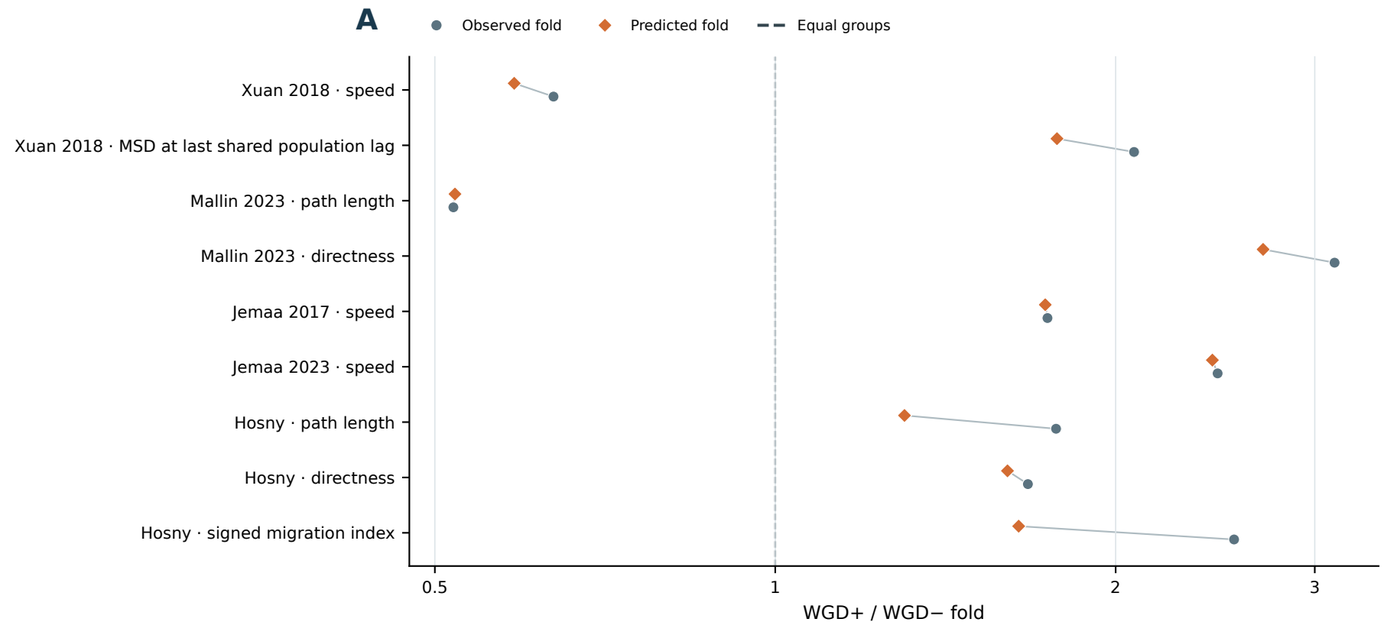
Complex spatial Fourier coefficient of active concentration u on a ring of physical circumference C; n is the integer mode index and q_n its physical wavenumber.

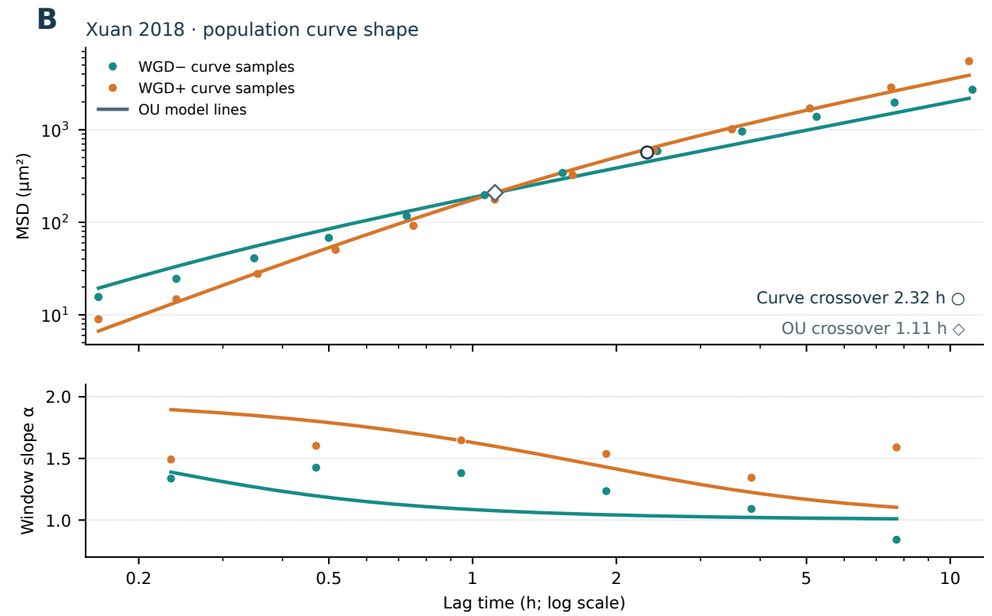
Sources and reproducibility

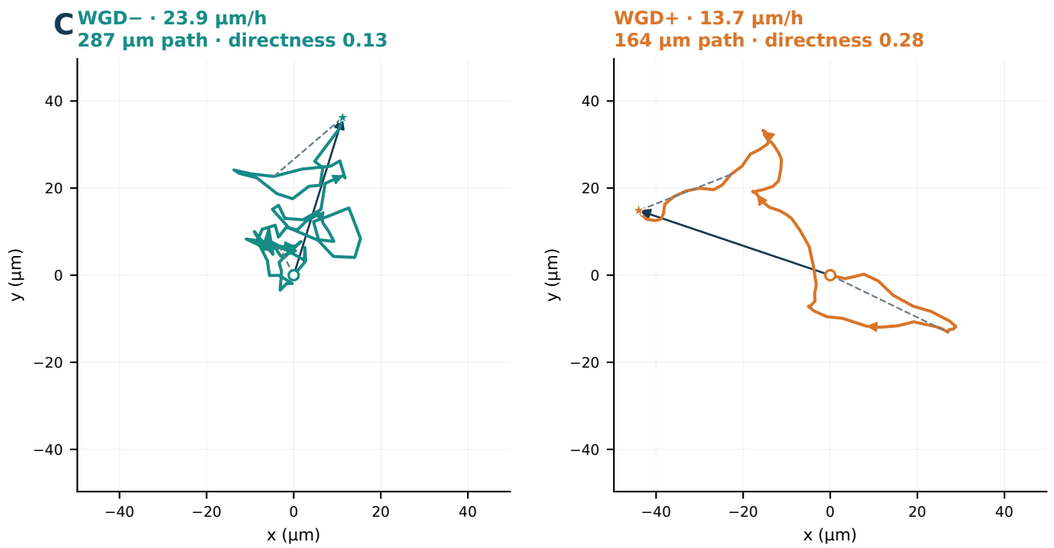
Main source data and definitions

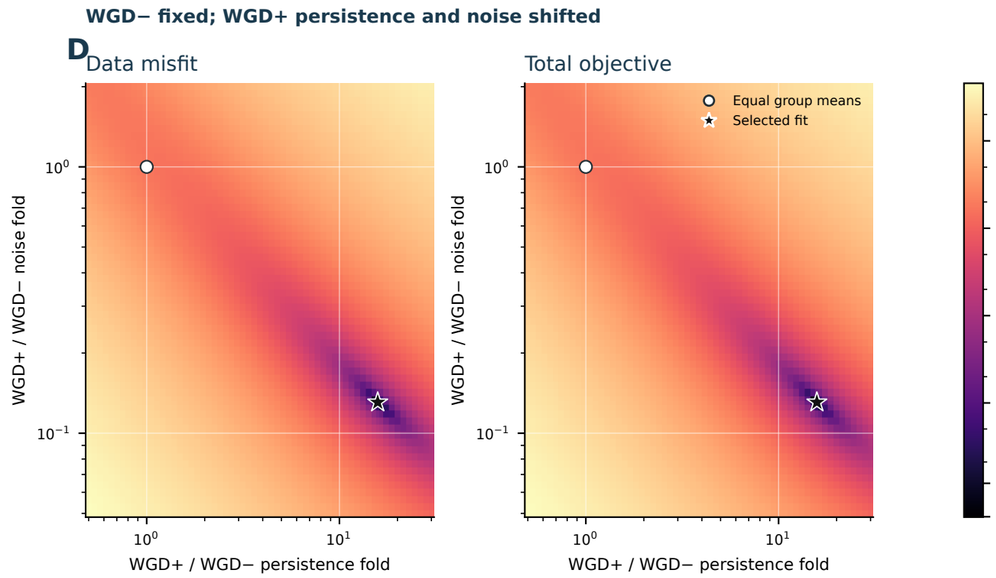
Exact source hashes and generation time

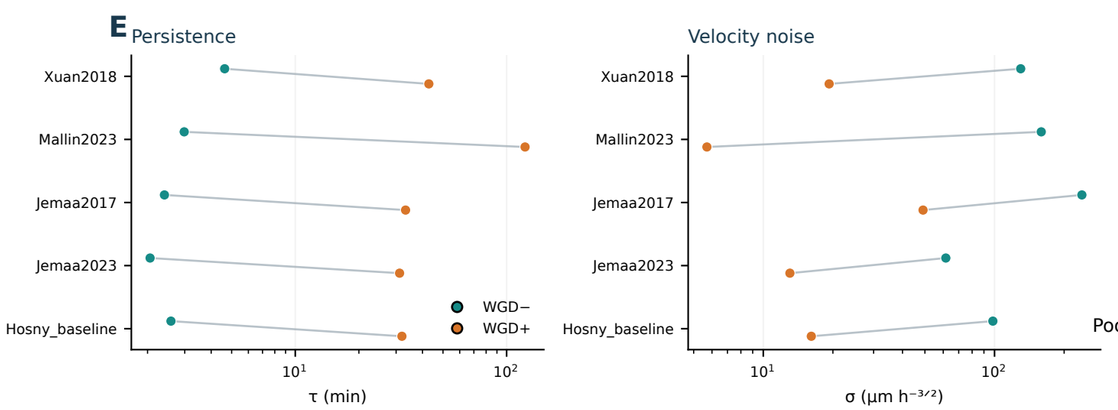
Figure 6 | Migration summaries and conditional OU dynamics

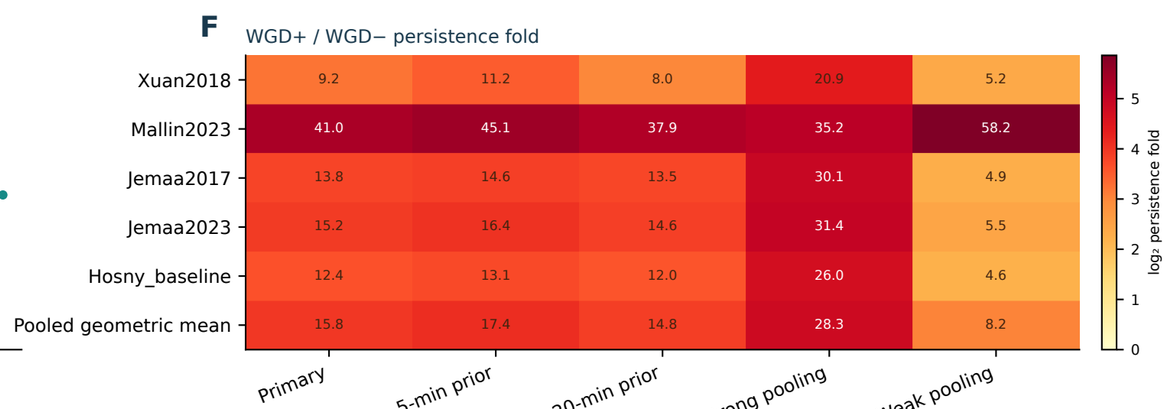










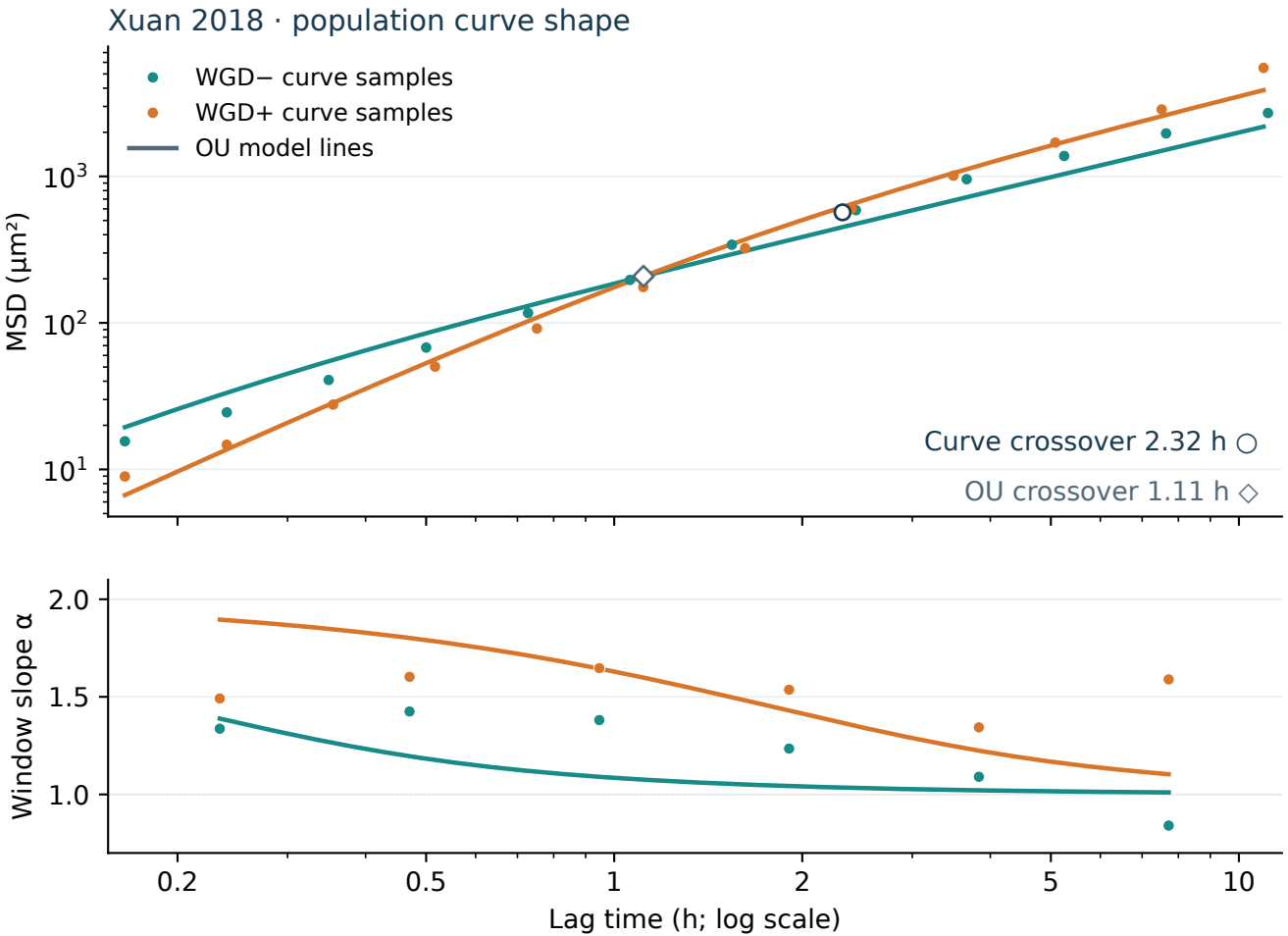


Points sample digitized published curves; original experimental lag grid unknown.

B slopes describe curve shape, not independent measurements or OU memory times. C tracks are simulations. D/F show conditional objective/prior sensitivity.

No new fit or simulation; A and C–F retain their original numerical evidence. See the enlarged B detail for readable curve-shape comparison.

Figure 6B | Digitized population MSD and curve shape



Points sample digitized published curves;
original experimental lag grid unknown.

Lower points: newly estimated window slopes. Correlated raster samples; no biological error bars.

Figure 6. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 6. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 6. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Migration display and preserved-input provenance
Migration display numerical and preservation checks

Figure 7. RESERVED: new WGD motility results

RESERVED MAJOR EXPERIMENTAL RESULTS

New WGD motility assays — all data pending

A Paired biological system

SUM-159 chemically induced WGD+/− pilot

Confirm group identity, size and morphology

PENDING — no observations available

B Imaging and observed tracks

Optimize acquisition and tracking

Declare cadence, duration, eligibility and replicates

PENDING — no observations available

C Measured migration behaviour

Speed, path length, displacement and directness

Lag-dependent MSD under declared observation rules

PENDING — no observations available

D Evaluate the persistence hypothesis

Relate the acquired observations to WGD and size

Retain null or opposite outcomes; specify controls

PENDING — no observations available

Proposed panel plan only. Design and analysis remain to be agreed before interpreting data.

Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 7. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 7. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification
Figure 2 October 20 revision brief
Figure 3 October 20 revision brief
Figure 4 October 20 revision brief
Round-7 brief
Target narrative; quantitative statements checked against the rebuilt evidence
Pilot methods and comparison values; independent reimplementation